

Satellite Realignment System

Operational Report — IMU Feedback + GPR Auto-tracking

Generated: 2026-05-05 09:53 UTC

1. System Overview

Parameter	Value
Stations Monitored	2
Online Stations	2
GPR Models Trained	0
IMU Feedback	Active (MPU-6050)
Architecture	ESP32 → FastAPI → Dashboard
Report Timestamp	2026-05-05 09:53 UTC

2. Station Status

Station	Status	Mode	AZ (°)	EL (°)	IMU AZ (°)	IMU EL (°)	Signal (dBm)	IMU Cal
Station A – Harare	ONLINE	Mode.AUTO	65.0	160.0	0.0	0.0	-17.0	No
Station B – Victoria Falls	ONLINE	Mode.AUTO	0.0	0.0	0.0	0.0	-21.0	No

3. Movement Verification Log

Time	Station	Result	IMU AZ Δ (°)	IMU EL Δ (°)	Target AZ	Target EL
2026-05-04T22:56	Station A – Harare	FAIL	7.91	0.00	0.0°	127.9°
2026-05-04T22:55	Station A – Harare	FAIL	35.94	0.00	0.0°	127.9°

4. Position vs Signal Log

Time	Station	AZ (°)	EL (°)	Signal (dBm)
2026-05-05T09:52	Station A – Harare	65.0	160.0	-17.0
2026-05-05T09:43	Station A – Harare	65.0	160.0	N/A
2026-05-05T09:42	Station A – Harare	65.0	160.0	N/A
2026-05-05T09:42	Station A – Harare	65.0	160.0	N/A
2026-05-05T09:42	Station A – Harare	65.0	160.0	N/A
2026-05-05T09:41	Station A – Harare	65.0	160.0	N/A
2026-05-05T09:29	Station A – Harare	65.0	160.0	N/A
2026-05-05T08:43	Station A – Harare	65.0	160.0	N/A
2026-05-05T08:43	Station A – Harare	65.0	160.0	N/A
2026-05-05T08:43	Station A – Harare	65.0	160.0	N/A

2026-05-05T08:42	Station A – Harare	65.0	160.0	N/A
2026-05-05T08:42	Station A – Harare	65.0	160.0	N/A
2026-05-05T08:40	Station A – Harare	65.0	160.0	N/A
2026-05-05T08:17	Station A – Harare	65.0	160.0	N/A
2026-05-05T08:17	Station A – Harare	65.0	160.0	N/A
2026-05-05T08:17	Station A – Harare	65.0	160.0	N/A
2026-05-05T08:17	Station A – Harare	65.0	160.0	N/A
2026-05-05T08:17	Station A – Harare	65.0	160.0	N/A
2026-05-05T08:16	Station A – Harare	65.0	160.0	N/A
2026-05-05T08:15	Station A – Harare	65.0	160.0	N/A
2026-05-05T08:15	Station A – Harare	65.0	160.0	N/A

5. GPR Signal Model

Station	Trained	Samples	Trained at	Kernel (summary)
Station A – Harare	No	—	—	—
Station B – Victoria Falls	No	—	—	—

Top predicted angles (10° grid)

Station	Rank	AZ (°)	EL (°)	Predicted (dBm)	Confidence (%)
No data	—	—	—	—	—

6. Sweep Status

Station	State	Progress (%)	Positions	Started at	Finished at
Station A – Harare	SweepState.COMPLETE	0.0	0 / 0	—	—
Station B – Victoria Falls	SweepState.IN_PROGRESS	0.0	0 / 25	2026-05-05T09:52	—

7. Environmental Data

Station	Timestamp	Temp (°C)	Wind (km/h)	Rain	Humidity	Pressure
Station_2	2026-05-05T09:53:00.009061+00:00	29.9	10.3	0.0	41%	918
Station_1	2026-05-05T09:52:58.590374+00:00	24.3	10.4	0.0	41%	860
Station_2	2026-05-05T09:49:02.570598+00:00	29.9	10.3	0.0	41%	918
Station_1	2026-05-05T09:49:02.558250+00:00	24.3	10.4	0.0	41%	860
Station_2	2026-05-05T09:48:49.141322+00:00	29.9	10.3	0.0	41%	918
Station_1	2026-05-05T09:48:47.808152+00:00	24.3	10.4	0.0	41%	860
Station_2	2026-05-05T09:44:03.557812+00:00	29.6	9.5	0.0	42%	918
Station_1	2026-05-05T09:44:03.536004+00:00	24.0	10.4	0.0	43%	860
Station_2	2026-05-05T09:43:46.328561+00:00	29.6	9.5	0.0	42%	918

2026-05-05T09:43:44.794534+00:00	Station_1	24.0	10.4	0.0	43%	860
2026-05-05T09:43:03.645240+00:00	Station_2	29.6	9.5	0.0	42%	918
2026-05-05T09:43:02.198630+00:00	Station_1	24.0	10.4	0.0	43%	860
2026-05-05T09:42:42.536005+00:00	Station_2	29.6	9.5	0.0	42%	918
2026-05-05T09:42:41.002802+00:00	Station_1	24.0	10.4	0.0	43%	860
2026-05-05T09:42:31.743929+00:00	Station_2	29.6	9.5	0.0	42%	918
2026-05-05T09:42:30.084277+00:00	Station_1	24.0	10.4	0.0	43%	860
2026-05-05T09:42:02.311021+00:00	Station_2	29.6	9.5	0.0	42%	918
2026-05-05T09:42:00.765174+00:00	Station_1	24.0	10.4	0.0	43%	860
2026-05-05T09:40:11.384129+00:00	Station_2	29.6	9.5	0.0	42%	918
2026-05-05T09:40:05.701864+00:00	Station_1	24.0	10.4	0.0	43%	860

8. System Logs (Last 30)

Timestamp	Station	Level	Message
2026-05-05T09:52	station_2	INFO	Cold-start sweep initiated
2026-05-05T09:52	system	INFO	Backend v4 started — IMU feedback + GPR auto-tracking active
2026-05-05T09:52	station_2	WARN	GPR not enough data yet: Need at least 8 samples to train. Have 0. Run a sweep first.
2026-05-05T09:52	station_2	INFO	Station initialised: station_2
2026-05-05T09:52	station_1	WARN	GPR not enough data yet: After filtering invalid readings, only 0 valid samples remain. Need 8.
2026-05-05T09:52	station_1	INFO	Station initialised: station_1
2026-05-05T09:43	system	INFO	Backend v3 started — IMU feedback active
2026-05-05T09:43	station_2	INFO	Station initialised: station_2
2026-05-05T09:43	station_1	INFO	Station initialised: station_1
2026-05-05T09:42	system	INFO	Backend v3 started — IMU feedback active
2026-05-05T09:42	station_2	INFO	Station initialised: station_2
2026-05-05T09:42	station_1	INFO	Station initialised: station_1
2026-05-05T09:42	system	INFO	Backend v3 started — IMU feedback active
2026-05-05T09:42	station_2	INFO	Station initialised: station_2
2026-05-05T09:42	station_1	INFO	Station initialised: station_1
2026-05-05T09:42	system	INFO	Backend v3 started — IMU feedback active
2026-05-05T09:42	station_2	INFO	Station initialised: station_2
2026-05-05T09:42	station_1	INFO	Station initialised: station_1
2026-05-05T09:42	system	INFO	Backend v3 started — IMU feedback active
2026-05-05T09:42	station_2	INFO	Station initialised: station_2
2026-05-05T09:42	station_1	INFO	Station initialised: station_1
2026-05-05T09:41	system	INFO	Backend v3 started — IMU feedback active
2026-05-05T09:41	station_2	INFO	Station initialised: station_2
2026-05-05T09:41	station_1	INFO	Station initialised: station_1
2026-05-05T09:41	system	INFO	Backend v3 started — IMU feedback active

2026-05-05T09:41	station_2	INFO	Station initialised: station_2
2026-05-05T09:41	station_1	INFO	Station initialised: station_1
2026-05-05T09:40	station_1	INFO	Mode changed to Mode.AUTO
2026-05-05T09:40	station_1	INFO	Mode changed to Mode.MANUAL
2026-05-05T09:29	system	INFO	Backend v3 started — IMU feedback active

End of Report — Satellite Realignment System v4 (GPR auto-tracking)